

Name:

Roll No.

Signature :

Sep 19, 2024 (Thu)

MS101 – Makerspace 2024-25/I (Autumn Semester)

EE Mid-semester Examination

Time: 2 hours

Marks: 50

1. This **Question-cum-Answer Booklet** has 8 pages and 18 questions.
2. Write your **answers only in the space provided for answers**. Answers written at any other place will not be checked. You may use the page margins also for rough work.
3. No explanations / clarifications will be given to any of the questions.
4. No negative marks for wrong answers, **however steps are required for all numerical answers.**
5. **In case of questions with multiple-choices, credit will be given only if all correct options are identified.**

Q.no.	1	2	3	4	5	6	7	8	9	10	11	12
Marks												
Q.no.	13	14	15	16	17	18					TOTAL	
Marks												

1. Write all the correct options out of the following statements regarding Digital Multimeter (DMM).
 - A) The input resistance of a DMM for voltage measurements is typically 10 k Ω .
 - B) It should not be used for measuring AC voltage if the frequency is much higher than 50 Hz or the waveform is non-sinusoidal.
 - C) It is set for ohm measurement and connected across a capacitor. The reading will be "1".
 - D) It is set for ohm measurement and connected across an inductor. The reading will be "1".
 - E) The instrument may be damaged if it is set for ohm measurement and connected across a voltage.

Answer(s):

Mark: 1

2. Write all the correct options out of the following statements on MS101 Lab Digital Storage Oscilloscope (DSO).
 - A) The input resistance of a DSO channel (without using the DSO probe) is 1 M Ω .
 - B) The input resistance of a DSO channel (without using the DSO probe) is 5 M Ω .
 - C) DSOs are versatile electronic instruments used for displaying and measuring time-varying current signals.
 - D) A differential signal across nodes A and B can be viewed on CH1 by connecting the CH1 signal input to node A and the GND to node B.
 - E) A differential signal across nodes A and B can be viewed on CH1 by connecting the CH1 signal input to node B and the GND to node A.
 - F) A differential signal across nodes A and B cannot be viewed on CH1 by connecting the CH1 signal input to node A and the GND to node B or by connecting the CH1 signal input to node B and the GND to node A.

Answer(s):

Mark: 1

Rough work

3. A practical current source is modelled as an ideal current source of I_S mA in parallel with a resistance of $80\text{ k}\Omega$. When a resistance of $4\text{ k}\Omega$ was connected across this source, the voltage across it was measured to be 20 V . What is the value of I_S in mA?

No marks without steps.

Marks: 2

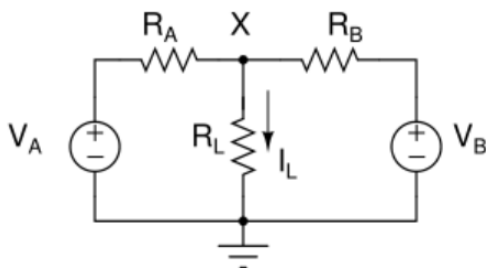
$I_S = \text{----- mA}$

4. A resistive network is given below. $V_A = 18\text{ V}$, $V_B = -9\text{ V}$, $R_A = R_B = R_L = 600\text{ }\Omega$.

Evaluate the node voltage V_X and the current I_L . Hint: Use superposition method. *No marks without steps.*

Marks: 3[=1.5+1.5]

Steps:

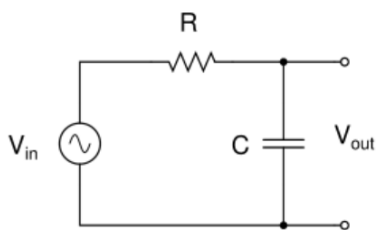


$V_X = \text{----- V}$; $I_L = \text{----- mA}$

5. An RC filter circuit is shown below. $R = 2\text{ k}\Omega$, $C = 0.02\text{ }\mu\text{F}$. Its gain A is given by:

$$A = \left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{[1+(\omega CR)^2]^{1/2}} = \frac{1}{[1+(f/f_c)^2]^{1/2}}$$

where $\omega = 2\pi f$ and $f_c = 1/(2\pi RC)$.



- A) Evaluate f_c in Hz.
B) Evaluate A for $f = f_c$.
C) Evaluate A for $f \ll f_c$.
D) Give a sketch of A as a function of f , indicating its values at f_c and $f \ll f_c$.

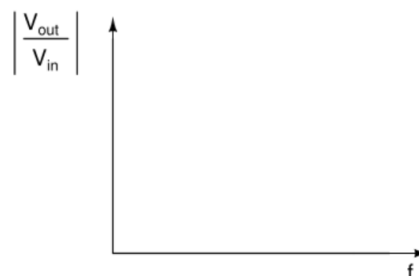
Marks: 3[=1+0.5+0.5+1]

A) $f_c = \text{----- Hz}$.

B) A for $f = f_c$:

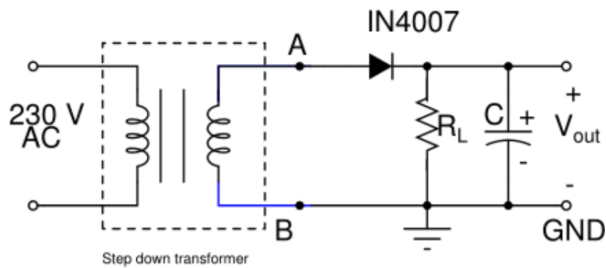
C) A for $f \ll f_c$:

D) Sketch of A as a function of f



Rough work:

6. The rectifier circuit shown below uses a 230 V (RMS) to 15 V (RMS) step-down transformer.



Marks: 3[=1+1+1]

A) Evaluate the average value of V_{out} if C is removed.

V_{out} (average) =

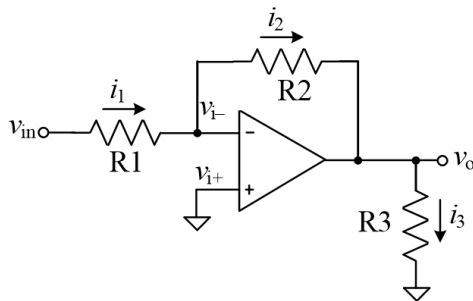
B) With the capacitor C present and the load resistor R_L not present or very large, evaluate V_{out} .

V_{out} =

C) Find the V_{out} as in (B) if the transformer input voltage is 115 V (RMS)

V_{out} =

7. In the op-amp circuit shown in the figure, $R_1 = 80 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$, $R_3 = 500 \Omega$. Find the voltage, current, and power gains. Is it serving as an amplifier?



Marks: 4 [= 1 x 4]

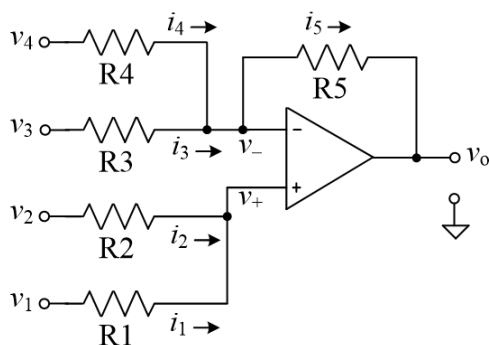
A) Voltage gain $A_v = v_o / v_{in}$

B) Current gain $A_i = i_3 / i_1$

C) Power gain $A_p = A_v A_i$

D) Is it serving as an amplifier? Write Yes or No.

8. In the summer circuit shown in the figure, $R_1 = R_2 = R_3 = 20 \text{ k}\Omega$, $R_4 = R_5 = 100 \text{ k}\Omega$. Find the voltage gain and input resistance for the second and third inputs.



Marks: 4 [=2+0.5+1+0.5]

A) $A_{v2} = v_o / v_2$

B) $R_{in2} = v_2 / i_2$

C) $A_{v3} = v_o / v_3$

D) $R_{in3} = v_3 / i_3$

Rough work:

9. In the 5-bit binary system with the MSB representing the negative sign, the range of decimal numbers that can be represented is:

Marks: 2

Minimum decimal number = ; Maximum decimal number =

10. Write the hexadecimal equivalent of the decimal number 481.

Mark: 1

Answer:

11. The truth table of a logical function (F_n) is given below. Obtain the minimized logic expression of the function using Karnaugh map. Use the given Karnaugh map format.

Marks: 4 [= 2+2]

No	A	B	C	D	F_n
00	0	0	0	0	1
01	0	0	0	1	0
02	0	0	1	0	0
03	0	0	1	1	0
04	0	1	0	0	1
05	0	1	0	1	1
06	0	1	1	0	1
07	0	1	1	1	1
08	1	0	0	0	1
09	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	0
15	1	1	1	1	0

AB↓ CD→	00	01	11	10
00				
01				
11				
10				

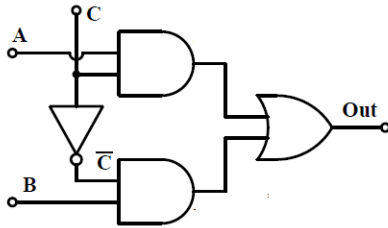
$F_n =$

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Rough work:

12. A digital circuit is shown below. Write the Boolean expression for **Out**.

Marks: 2



Out =

13. An Analog-to-Digital Converter (ADC) is used to obtain the digital value (represented value) corresponding to the nearest level which is less than the analog voltage. The input voltage varies from 0 to 5 V. What will be the ADC output in the binary form for the input voltage of 2.75 V if the ADC is:

- A) a 3-bit ADC
- B) a 6-bit ADC

Steps:

Marks: 4[=2+2]

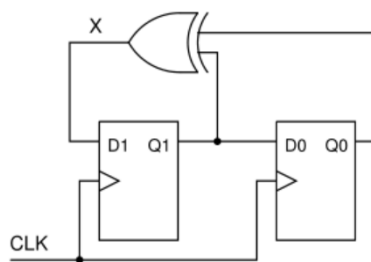
A) -----

;

B) -----

14. A sequential digital circuit is shown below. Assume that initially **Q1 = 0**, **Q0 = 1**, and consequently the XOR output **X = 1**. Determine the values of **Q1**, **Q0** and **X** for four **CLK** pulses. Enter the values after each **CLK** in the table shown. The first row is filled for the initial case.

Marks: 4[=4x1]

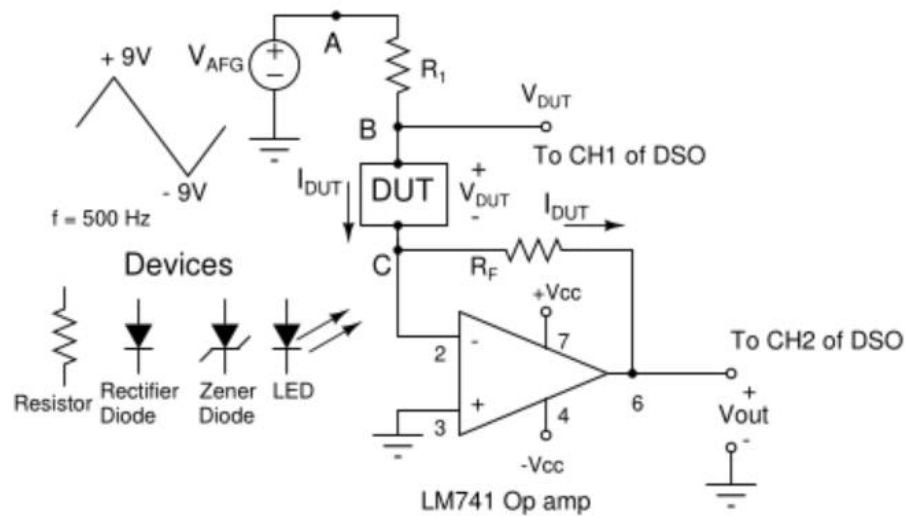


CLK	Q1	Q0	X
0	0	1	1
1			
2			
3			
4			

Rough work:

15. Op amp based current-to-voltage converter circuit of Expt 2 is shown below. Given: $R_1 = 2\text{ k}\Omega$, $R_F = 2\text{ k}\Omega$, $+V_{cc} = +12\text{ V}$, $-V_{cc} = -12\text{ V}$. A Zener diode is connected as the device-under-test (DUT) with its anode at terminal B and cathode at terminal C. The Zener voltage is 4.5 V . Calculate the current through the Zener diode (I_{DUT}), V_{DUT} and V_{out} at the instant when $V_{AFG} = -8\text{ V}$.

Marks: 5[=2+2+1]



Show steps:

$I_{DUT} =$ mA. $V_{DUT} =$ V. $V_{out} =$ V.

Rough work:

16. Arduino library provides the “map” function to map a value from one given range to another. It is called as:
newValue = map (oldValue, fromRangelow, fromRangehigh, toRangelow, toRangehigh).

What will be the returned value if the map function is called with the following arguments:

A) 800, 0, 1023, 0, 255

B) 110, 0, 511, 127, 0

Mark: 3[=1.5+1.5]

Steps:

A) ; B)

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Rough work:

17. The Arduino function **analogWrite(3, 64)** will output a PWM waveform on pin 3 of Arduino Nano/Uno.

- A) Sketch the waveform you will observe on pin 3. You may assume that the period of the PWM waveform is 2 ms.
- B) Evaluate the average value of the above waveform.

Note: PWM waveform maximum value = 5 V; PWM waveform minimum value = 0 V.

Mark: 3[=1.5+1.5]

18. Write all the correct options out of the following statements regarding microprocessors and associated circuits.

- A) The program counter is a register which is also referred to as the data pointer.
- B) The counter which provides addresses to the program memory is called a program counter.
- C) The output of tri-stateable digital circuits can have three states, viz. '0' or '1' or a low impedance state.
- D) The output of tri-stateable digital circuits can have three states, viz. '0' or '1' or a high impedance state.
- E) Moving to a different block of instructions may be accomplished by loading a different value into the program counter.
- F) Moving to a different block of instructions may be accomplished by loading a different value into data pointer.

Answer(s):

Mark: 1

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Rough work: